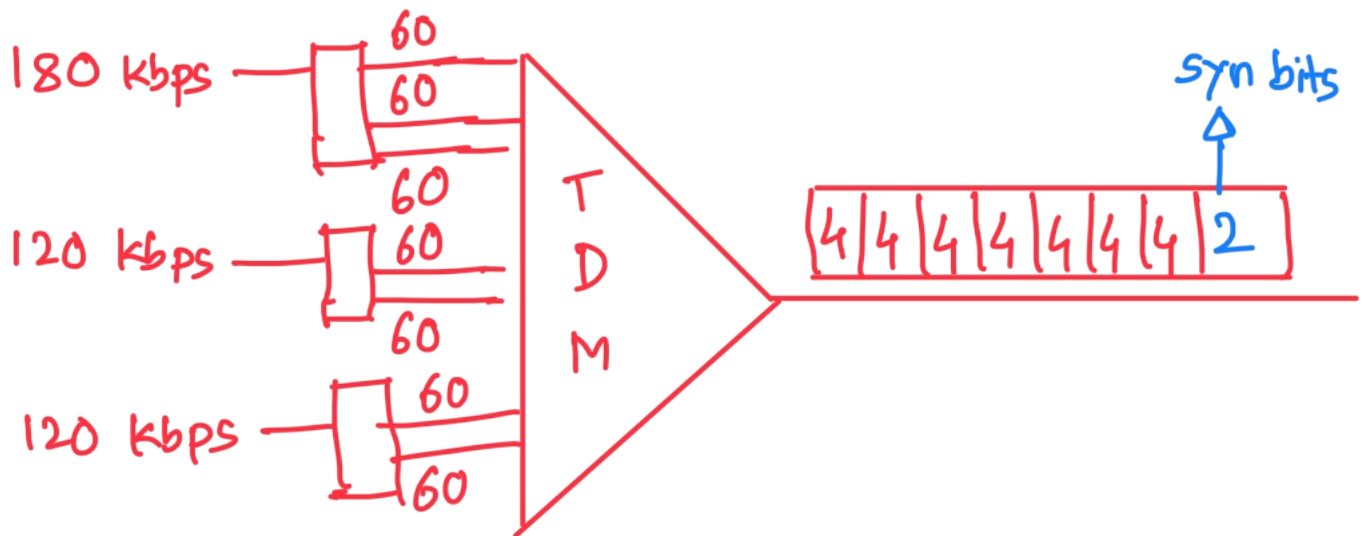


Name:		ID:		Section:	
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1. Three channels, ONE with a bit rate of 180 kbps and TWO with a bit rate of 120 kbps, are to be multiplexed, with TWO synchronization bits and FOUR interleaved bits. You cannot use pulse stuffing.

- Show how this can be achieved (figure only). [3]
- What will be the frame size, frame rate & frame duration? [3]
- What will be the output rate and output slot duration? [2]



$$\text{frame size} = 7 \times 4 + 2 = 30 \text{ bits}$$

$$\text{frame rate} = \frac{60 \times 10^3}{4} = 15,000 \text{ fps}$$

$$\text{frame duration} = \frac{1}{15000} = 6.67 \times 10^{-5} \text{ s}$$

$$\begin{aligned} \text{o/p bit rate} &= \text{frame size} \times \text{frame rate} \\ &= 450,000 \text{ bps} = 450 \text{ kbps} \end{aligned}$$

$$\text{o/p duration} = \frac{1}{\text{o/p bit rate}} = 2.22 \times 10^{-6} \text{ s}$$

$$\therefore \text{o/p slot duration} = \text{unit} \times \text{o/p bit duration} = 8.89 \times 10^{-6} \text{ s}$$

$$\text{Note: o/p slot dur} = \frac{\text{frame duration}}{\text{unit}} = 1.67 \times 10^{-5} \text{ s} \quad \text{ALSO correct (conceptually)}$$

2. An error detection and correction system uses CRC with the generator **1011**. Under this scheme, a receiver received the codeword **110101101** from a sender. $\rightarrow$ 9 bits

- Identify the length of the dataword and write the scheme in $C(n, k)$ format. [2]
- How many codewords are possible in this scheme? How many of them are invalid? [2]
- Determine whether the received codeword contains any error or not. Show full calculations. [3]

Generator 4 bits $\rightarrow$ Remainder 3 bits.

Data word $\leftarrow$ 110101101 $\rightarrow$ Remainder

$\therefore n = 9$ bits, $k = 6$ bits $\rightarrow C(9, 6)$

Total codewords = $2^9 = 512$, valid = $2^6 = 64$

$\therefore$ Invalid = $512 - 64 = 448$ codewords

C. $1011 \overline{) 110101101}$

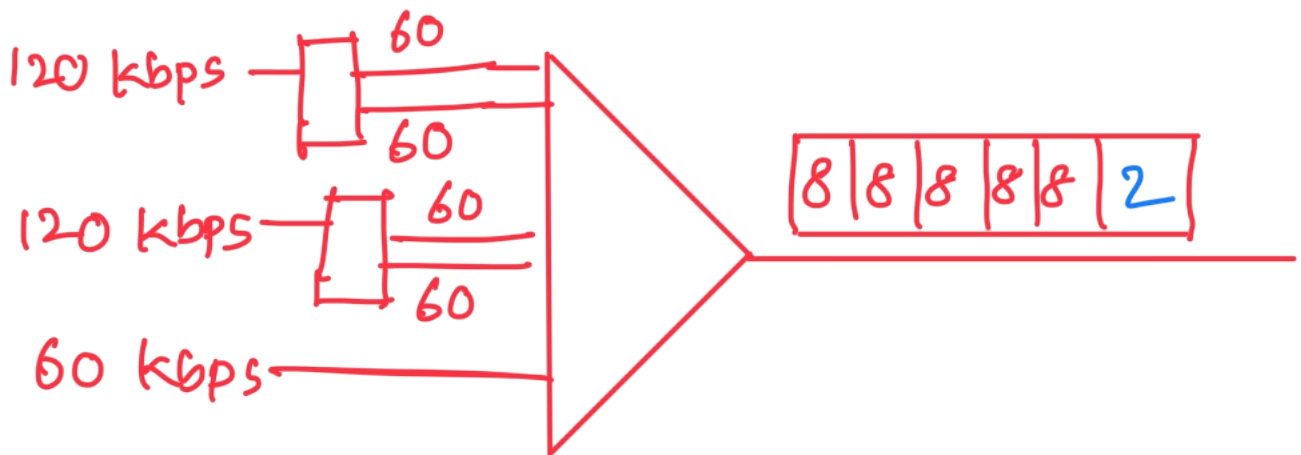
$$\begin{array}{r}
 1011 \overline{) 110101101} \\
 \underline{1011} \\
 1100 \\
 1011 \\
 \underline{1111} \\
 1011 \\
 \underline{1001} \\
 1011 \\
 \underline{001001} \\
 1011 \\
 \underline{0010}
 \end{array}$$

Remainder $\neq 0 \rightarrow$ Error.

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1. **Three** channels, **TWO** with a bit rate of **120 kbps** and **ONE** with a bit rate of **60 kbps**, are to be multiplexed, with **TWO** synchronization bits and **EIGHT** interleaved bits (multiplexing unit). You cannot use pulse stuffing.

- Show how this can be achieved (figure only). [3]
- What will be the frame size, frame rate & frame duration? [3]
- What will be the output rate and output slot duration? [2]



$$\text{frame size} = 5 \times 8 + 2 = 42 \text{ bits}$$

$$\text{frame rate} = \frac{60 \times 10^3}{8} = 7,500 \text{ fps}$$

$$\text{o/p bit rate} = 42 \times 7500 = 315000 \text{ bps}$$

$$\text{o/p bit duration} = \frac{1}{315000} = 3.17 \times 10^{-6} \text{ s}$$

$$\therefore \text{o/p slot durc.} = \frac{\text{o/p bit dur.}}{\text{unit}} \times \text{unit} = 2.54 \times 10^{-5} \text{ s}$$

$$(\text{OR}) = \frac{\text{f. duration}}{\text{unit}} = 1.66 \times 10^{-5} \text{ s}$$

$$\text{frame duration} = \frac{1}{7500} = 1.33 \times 10^{-4} \text{ s.}$$

2. An error detection and correction system uses CRC with the generator **11001**. Under this scheme, a receiver received the codeword **101110101** from a sender. $n = 11$ bits

- Identify the length of the dataword and write the scheme in $C(n, k)$ format. [2]
- How many codewords are possible in this scheme? How many of them are invalid? [2]
- Determine whether the received codeword contains any error or not. Show full calculations. [3]

Generator 5 bits $\rightarrow$ remainder 4 bits.

D/w $\leftarrow$ 1 0 1 1 1 0 1 $\rightarrow$ rem 0 1 0 1 $C(11, 7)$
7 bits

total codewords = $2^{11} = 2048$, valid = $2^7 = 128$

invalid codewords = $2048 - 128 = 1920$

C. 11001 $\left| \begin{array}{r} 101110101 \\ 11001 \downarrow \\ \hline \text{Ø} 11100 \\ 11001 \downarrow \\ \hline \text{ØØ} 10110 \\ 11001 \downarrow \\ \hline \text{Ø} 11111 \\ 11001 \downarrow \\ \hline \text{ØØ} 11001 \\ 11001 \downarrow \\ \hline 00000 \end{array} \right.$

Rem = 0 $\rightarrow$ No ERROR.